

Project 3.2

:Essential Requirements

RFM / Price History

Columns

Customer

- customer ID •
- name •
- age •
- email •
- signup date (= registration date) •
- country •
- marketing opt-in (true / false) •

Product

- product ID •
- category •
- name •
- price used — سعر المنتج للعميل •
- cost used — التكلفة الأصلية •
- margin — الربح الصافي من القطعة •

Order

- order ID •
- customer ID •
- order time •
- payment method •
- discount PCT •
- subtotal usd — السعر قبل إضافة الـ discount •
- total usd — السعر بعد إضافة الـ discount •
- device •
- country •
- source — مصدر الزيارة للطلب •

Order Items

- order ID •
- product ID •
- unit price used •
- quantity •

- line total used — unit price × quantity

Schema / Shape

Table	Rows	Columns
Customer	20,000	7
Product	1197	6
Order	33580	10
Order Items	59163	5

Dataset Notes On type

- أي عمود date في dataset، نوعه object مش date.

Nulls

- مفيش nulls.

Duplicates

- فيه duplicates في order items بس.

Invalid Values

- customer: مفيش.
- orders: مفيش.
- product: فيه invalid في margin used — مش بتساوي price used – cost used.
- order items: فيه invalid في line total used — مش بتساوي quantity × unit price.

Primary Keys

- customer: customer ID
- order: order ID
- product: product ID
- order items: order ID + product ID (composite)

Foreign Keys

- customer: مفيش
- order: customer ID
- product: مفيش
- order items: order ID, product ID

Relations

- customer ↔ order — one to many •
- order ↔ order items — one to many •
- product ↔ order items — one to many •

Grain

- customer:** one row per customer •
- order:** one row per order •
- product:** one row per product •
- order items:** one row per product within an order •

Important Columns

For RFM

Tables: customer, order

- :customer** •
- customer ID •
- :order** •
- ,customer ID •
- ,order ID (→ frequency) •
- order time (→ recency) •
- Total usd (→ monetary) •

For Price History

Tables: products, order items, orders

- product: product ID, price used •
- order items: product ID, unit price •
- orders: order ID, order time •

Generated Columns

- :RFM** •
- recency, frequency, monetary, RFM score •
- :Price History** •
- product price, data warehouse start price, data warehouse end price •